P515/1
Principles and Practice of
Agriculture
July/August, 2026
3 hours

MUKONO EXAMINATION COUNCIL 2026

Uganda Advanced Certificate of Education

Principles and Practice of Agriculture

P515/1

THEORY

3 hours

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of two sections; **A** and **B**. It has **six** items.*

*Section **A** has two **compulsory** items.*

*Section **B** has two parts; **I** and **II**. Respond to **one** item from **each** part.*

Respond to four examination items in all.

*Any additional item(s) responded to will **not** be scored.*

***All responses must** be written in the answer booklet(s) provided.*

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SECTION A

Value addition for the market and basic science

Item 1

Mukwasi is an agro-produce dealer who supplies fruits such as oranges, mangoes, passion fruits, pawpaws, and watermelons. He sells mangoes at Ushs 10,000 per basket, oranges at Ushs 15,000 per basket, passion fruits at Ushs 20,000 per basket, pawpaws at Ushs 3,500 per fruit, and watermelons at Ushs 5,500 per fruit.

The fruits are transported on a wooden cart that is manually pushed by his marketing agents. His sales are mainly chance-based. Customers who buy passion fruits often complain that large and small fruits are mixed together, including some that are still raw. Within his area of operation, there are many small-scale producers of similar fruits who sell directly to consumers at producer prices. Unsold fruits are always stored overnight in a room that also houses local chickens. Over time, Mukwasi's daily sales declined rapidly. One of his regular customers purchased 10 baskets of mangoes, 7 baskets of oranges, 4 baskets of passion fruits, 15 pawpaws, and 17 watermelons. She paid in cash and thereafter demanded proof of payment.

Task

a) Prepare the document that Mukwasi would give to his customer as proof of payment for the bought items.

b) Explain how Mukwasi can improve on the sales of his fruits.

Item 2

Justine is a poultry farmer who stocked his farm with 20 mature cocks. Five of the cocks have a tight and contracted cloaca. Some of the mature hens have small, firm, and less pliable abdomens. Upon candling fresh eggs from the flock, several show numerous dark spots. To reduce feeding costs, he feeds some pullets on fresh cassava. When the eggs are consumed, consumers often complain about a drug-like smell. Despite proper feeding, several hens also lay eggs with thin and fragile shells. Justine later realized that the productivity levels on his farm were lower than his expectation.

Task

a) Explain the causes of low productivity levels on Justine's poultry farm.

b) How should Justine improve on the productivity of his poultry farm?

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SECTION B
PART I ANIMAL PRODUCTION

Attempt one item from this part

Item 3

Namata established a dairy farm on an 8-hectare piece of open land, which she intends to fence using a four-strand barbed wire fence. She purchased eight rolls of barbed wire, although she is unsure whether this quantity will be sufficient for effective fencing. She stocked the farm with 34 animals. She constructed a dip tank for controlling external parasites; however, the level of acaricide in the dip tank keeps reducing each day despite the structure being well sheltered. Animals often crowd around the feeding troughs.

Heifers that are about to calve are housed together with bulls. For vaccination, Namata uses the same needle on all animals. She identifies animals by heating a branding iron until it is blue-hot and pressing it onto the selected body part for five minutes to create a permanent mark. All animals on the farm are given the same deworming dosage. While milking, she lubricates the teats by dipping her fingers into milk. At times, she milks the animals inside the cattle shed. Customers often visit the shed while milking is in progress to purchase milk. Namata consequently realized that she was obtaining low yields on her dairy farm.

Task

a) Explain the effects of the practices on Namata's farm could have led to low yields.
b) How should Namata improve on the productivity of her dairy farm?

Item 4

Kamyu is a rabbit farmer. He constructed a rabbit house in such a way that urine collects in one corner. Young and mature rabbits are housed together in the same compartments. Rabbits that are about to deliver usually dig tunnels through the floor. The cages have walls that extend up to the roof and only a small single door.

Some mature rabbits have long claws. During vaccination, he vaccinates all rabbits, including those under treatment, to quicken their recovery. He castrates mature males using a burdizzo to crush the testes. Quite often, the rabbits cough and breathe with difficulty. He usually slaughters only fully fed rabbits and skins them two hours

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after slaughter. During evisceration, he opens the gut while it is still intact inside the carcass in order to clean it. Kamyu's records indicate that the yield on his farm is very poor.

Task

a) Explain the causes of low yields on Kamyu's farm.

b) Suggest with reasons how Kamyu can improve on the yield of his rabbit enterprise.

PART II
CROP PRODUCTION

Attempt one item from this part

Item 5

Chemutai grows wheat on five fields located in different areas on the slopes of Mount Elgon. She collaborates with the National Agricultural Research Organization (NARO) wheat improvement programme, which provides improved wheat varieties, soil testing services, and agronomic advice. Soil test results from her fields are shown in the table below.

Location	Soil depth (cm)	Soil pH	Yield (t/ha)
1	18	5.4	2.9
2	22	5.8	3.7
3	20	6.6	3.1
4	24	6.8	4.3
5	18	5.5	2.8

In one of the fields, she broadcast seeds together with ammonium sulphate fertilizer five weeks after the onset of rains, while in two other fields some seedlings were found buried in the lower parts of the field. She prepared the formerly grazed land using a single surface tillage operation, similar to the other fields. Some plants developed tunnel-like damage in their stems, while others showed stunted growth with yellowing leaves. During herbicide application, one worker was observed unblocking spray nozzles with his mouth. She harvested the crop from one of the fields immediately after the rains by stripping grains from the heads. She has observed variation in yield across her fields, although she remains committed to continuing the enterprise.

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Task

a) Explain the causes of variations in yield on Chemutai's fields.

b) How can Chemutai ensure high and stable productivity levels of her wheat fields.

Item 6

Angelina is a farmer of African egg plants. She partners with Nabuin Zonal Agricultural Research and Development Institute (Nabuin ZARDI) in Moroto District, which supplies her with improved seed and fertilizers, tests her soil, and compares the results with crop yield as shown in the table below.

Season	Organic matter (%)	Soil moisture (%)	Soil air (%)	Yield (kg/ha)
1	2.0	18	22	8.5
2	2.5	22	18	11
3	2.2	17	23	9
4	2.8	24	16	13

Five days after germination, her nursery bed was found to be fully mulched, while some nursery beds had experienced rapid death of seedlings. During transplanting, she placed fresh goat manure in the planting holes and immediately transplanted the seedlings. In one section of the garden, seedlings were transplanted after being hand-pulled from the nursery bed.

In the field, some plants exhibited stunted growth and yellowing of leaves, while others had their bark eaten off. In certain sections, the plants developed a bushy top. To avoid the inconvenience of carrying spraying equipment home, she always kept it hidden in the garden. Trees within the garden were left to grow with thick branches to provide shade for the eggplants. At harvest, she collected ripe fruits by breaking heavily loaded branches. Consequently, low yields are obtained on Angelina's farm.

Task

a) Explain the causes of low productivity on Angelina's African eggplant production enterprise.

b) Suggest with reasons how Angelina can improve on the of his African eggplant enterprise.

END

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